Next we learn

- Rank of a matrix
- Linear independence, dependence
- Vector spaces

Observation

Glyn James' book is not a very well written one.

We will continue our study of Matrix Theory from Kreyszig.

vectors: $\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}$

$$= \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1\hat{i} + 2\hat{j} + 3\hat{k}$$

Linear combination of vectors:

$$v_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$v = 3v_1 + 2v_2 = 3\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + 2\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$= \begin{bmatrix} 3x_1 \\ 3y_1 \\ 3z_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 2y_2 \\ 2z_2 \end{bmatrix} = \begin{bmatrix} 3x_1 + 2x_2 \\ 3y_1 + 2y_2 \\ 3z_1 + 2z_2 \end{bmatrix}$$

Linear combination of vectors:

$$v_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$2v_1 + 5v_2 + \begin{bmatrix} 6 \\ 1 \end{bmatrix}$$

not a linear combination of v_1, v_2

Linear combination of vectors:

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Dependent vectors:

A collection of vectors is called dependent if at least one of them can be written as a linear combination of other vectors.

e.g $\left\{ \underset{v_1}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}, \underset{v_2}{\begin{bmatrix} 2 \\ 0 \end{bmatrix}} \right\}$ $v_2 = 2v_1$ dependent

e.g. $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ not dependent

e.g $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$ dependent

$\because \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Assume: $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \vec{0} \implies \alpha_1 = \alpha_2 = \alpha_3 = 0$

Let if possible, $\{v_1, v_2, v_3\}$ be dependent

WLOG:

$$v_1 = \beta_1 v_2 + \beta_3 v_3 \quad \text{———} \quad (*)$$

$$\underline{1} v_1 - \beta_1 v_2 - \beta_3 v_3 = \vec{0}$$

Contradiction: $\because 1 \neq 0$.

Prove independence:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Independent if

If $\alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution $\alpha_1 = \alpha_2 = \alpha_3 = 0$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{——} \quad (*)$$

Show that $(*)$ has only $(0,0,0)$ as solution

: $\{v_1, v_2, v_3, v_4, \ldots, v_n\}$ is independent

if the equation

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = \vec{0}$$

has a unique solution

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = \vec{0}$$

Inependent vectors:

A collection of vectors is called independent if it is not dependent.

$\{v_1, v_2, v_3\}$ is independent

i.e. none of the vectors in the collection can be written as linear combination of others.

Theorem: $\{v_1, v_2, v_3\}$ is independent if and only if

the only solution for the equation

$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = \vec{0}$ is $\alpha_1 = \alpha_2 = \alpha_3 = 0$

Rank of matrix

- Rank of a matrix is the maximum number of linearly independent rows in it.
- It is same as the maximum number of independent column
- It is same as the number of pivots in its row reduced form.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{rank}(A) = 1$$

Example:

$$A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix} \rightarrow \text{find in matlab}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{rank}(A) = 1$$

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Today :

→ Vector space

→ Linear Transformation

Rank of matrix

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$Ax = b$

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Inependent vectors:

A collection of vectors is called independent if it is not dependent.

$\{v_1, v_2, v_3\}$ is independent

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

check that this is an independent collection of vectors:

Rank of matrix

- Rank of a matrix is the maximum number of linearly independent rows in it.
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Example:

$$A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

Theorem: Row equivalent matrices have the same ranks.

Vector Space

Main ingredients

- A set → elements will be called vectors
- Elements of set can be added together
- A collection of scalars → elements will be called scalars
- Scalars can be multiplied with vectors to produce vectors
- Addition and scalar multiplication has to satisfy some basic rules

V : set of vectors

$\boxed{\text{S: Set of scalars}}$

$\alpha, \beta \in S$

(a) $\boldsymbol{a} + \boldsymbol{b} = \boldsymbol{b} + \boldsymbol{a}$

(b) $\boldsymbol{a} + (\boldsymbol{b} + \boldsymbol{c}) = (\boldsymbol{a} + \boldsymbol{b}) + \boldsymbol{c}$

(c) there exists a zero vector $\mathbf{0}$ such that

$\boldsymbol{a} + \mathbf{0} = \boldsymbol{a}$

(d) for each $\boldsymbol{a}$ in V there is an element $-\boldsymbol{a}$ in V such that

$\boldsymbol{a} + (-\boldsymbol{a}) = \mathbf{0}$

V is a group under the binary operation "+".

(e) $\alpha(\boldsymbol{a} + \boldsymbol{b}) = \alpha\boldsymbol{a} + \alpha\boldsymbol{b}$ ✓

(f) $(\alpha + \beta)\boldsymbol{a} = \alpha\boldsymbol{a} + \beta\boldsymbol{a}$

(g) $(\alpha\beta)\boldsymbol{a} = \alpha(\beta\boldsymbol{a})$

(h) $1\boldsymbol{a} = \boldsymbol{a}$

Q. Check that the standard plane or 3d space is a vector space over real scalars.

$$V = \left\{ x\hat{i} + y\hat{j} + z\hat{k} \;\middle|\; x, y, z \in \mathbb{R} \right\}$$

"+" : standard vector addition

$\alpha \vec{v}$: standard scalar multiplication

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Q. Is the set of all polynomials a vector space over $\underbrace{\text{reals}}_{\text{scalars}}$?

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$$\vec{a} = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

$$\vec{b} = b_0 + b_1 x + b_2 x^2 + \cdots + b_m x^m$$

$$\vec{a} + \vec{b} = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + \cdots$$

Q. Is the set of all polynomials of at max deg 5 a vector space over reals?

Yes

Q. Is the set of all matrix a vector space?

→ no

sum of two matrix of different order is not defined

Q. Collection of all 2×3 real matrix.

Does this form a vector space ??

$$\alpha \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$= \begin{bmatrix} \alpha a_{11} & \alpha a_{12} & \alpha a_{13} \\ \alpha a_{21} & \alpha a_{22} & \alpha a_{23} \end{bmatrix}$$

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